Seshank Irukulapati

Email: seshanki25@gmail.com
Linkedin: linkedin.com/in/seshank-irukulapati-9a8657225
Mobile: +918951089312
Github: github.com/Seshank-I

Education

- **PES University (PESIT)** — Bengaluru, India
  Bachelor of Technology - Computer Science; CGPA: 8.15 — *August 2019 - July 2023*
  Courses: *Operating Systems, Data Structures, Analysis Of Algorithms, Networking, Databases*

Skills Summary

- **Languages:** Java, JavaScript, Python, SQL, Bash
- **Frameworks:** Spring Boot, Angular
- **Big Data:** Apache Kafka, Apache Spark, Apache Hive, Apache Trino, AWS S3, MinIO, Azure Blob, GCS Blobs
- **Tools:** PostgreSQL, Elasticsearch, Apache Ignite, MongoDB, Docker, Kubernetes, Git, Gradle

Experience

- **Subex Limited**
  Senior Software Engineer — *October 2024 - present*
  - Engineered a high-performance match system in **Java** using Tries. Developed a custom **cache** implementation with scheduled cache updates, and used a **Spark** UDF for distributed processing. Achieved throughput of **3000+** records/second/core.
  - Revamped case processing module using Kafka consumer pipeline with parallel processing, transactional retries, offset tracking in PostgreSQL database. Certified to process **1.2 million** cases per day.
  - Orchestrated seamless migration of **30TB** of customer data across phases with limited downtime, implementing fault tolerance and rollback mechanisms.
  - Optimized Spark streaming pipelines through salting, dynamic allocation, and phased aggregations — reducing runtime and memory usage by **20%.**
  - Led production stabilization efforts, resolving **100+** critical issues and delivered key optimizations improving performance and system reliability, while maintaining SLA compliance.
  - **Skills:** Java, Spring Boot, Kafka, Spark, Ignite, PostgreSQL, Multi-threading, Distributed Systems
- **Subex Limited**
  Software Engineer — *August 2023 - October 2024*
  - Implemented multi-cloud partitioning and backup strategy across S3, Azure Blob, GCS Blobs, PostgreSQL and Elasticsearch. Created data cleanup policy which can be used for any table across the platform. Reduced redundant data by **30%**, significantly reducing storage costs and manual intervention.
  - Improved the performance of telemetry system by **60%**, reducing data gathering time from **3 minutes to 50 seconds** by parallelizing the API calls to multiple microservices, refactoring the code and optimizing the SQL queries in dependent microservices.
  - Designed and implemented a **Kafka**-based task execution framework enabling real-time tracking across all application modules. Built a 30-day audit ledger for task re-triggering and audit trails.
  - Developed a DAG Rebuilder to reconstruct and optimize pipeline execution graphs, eliminating redundant node processing and improving overall performance by **40%**.
  - Implemented a configurable Spark-based summarization framework supporting dynamic filters, group-by dimensions, aggregation functions, and time-window logic for **Fraud Management**.
  - Developed and optimized RESTful Web Services in a microservice architecture using Java, Spring Boot and Hibernate, ensuring efficient data access and management.
  - **Skills:** Java, Spring boot, Spark, Kafka, PostgreSQL, Docker, Jenkins, Data lakes

Projects

- **Twitter Trends Analysis:** Developed a real-time Twitter stream processing pipeline , Integrated Twitter API to stream live tweets using Spark-Structured streaming and produced them to Kafka topics and used MongoDB to consume, store and categorize tweets, enabling efficient aggregation and analysis. Tech: Python, Spark, Kafka, MongoDB

Honors and Awards

- Collaborative Concorde Award: Team award for outstanding contribution to product engineering and innovation, 2024
- Placed 6th for our project on analyzing green cover in cities at CMRIT Hackathon, 2019
- Honoured with Spot Award (twice) for my contributions to the product releases, 2024